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EDUCATION

Master of Biomedical Science (Neuroscience)
University of Western Australia
Global Excellence Scholarship
*Master's Thesis with **High Distinction***

B.Tech, Biological Sciences & Bioengineering
IIT Kanpur
Silver Medal for Academic Excellence

GENAI & ML TRAINING

Product Management (Gen & Agentic AI)

BITS School of Management

- 6-month postgrad programme covering prompt engineering, RAG, LLMOps, agentic frameworks
- Capstone project – AI-powered Agent delivering curated company insights to sales reps before meetings to reduce manual research time and boost conversion.

Computational Neuroscience & Machine Learning

Neuromatch Academy

- Intensive 3-week Python course: Bayesian modelling, deep learning (CNNs, RNNs), reinforcement learning, systems neuroscience
- Invited back as Teaching Assistant in 2021 to mentor 9 students

SKILLS

Domain: Neuroimaging, BCI, computational psychiatry, genomics

Languages: Python, MATLAB, R, SQL, SPSS

Deep Learning: PyTorch, TensorFlow, CNN, RNN, Transformers

Data: fMRI, MRI, EEG, Confocal, Behavioural

Scientific Python: NumPy, pandas, scikit-learn, MNE-Python, Matplotlib

Deployment: Docker, FastAPI, CI/CD, Git, React

AI: RAG, LLMs, MCP, Agentic, Generative, LangChain, LangGraph

Yatin Azad

Computational Biologist | Neuroscience | Genomics

SUMMARY

“The researcher who ships.” Computational Neuroscientist with 4+ years building probabilistic models and analytical pipelines for neuroimaging and behavioural data. I catch critical data quality issues before they percolate into downstream modelling, improve model generalisation, and translate research pipelines into deployable tools.

RESEARCH EXPERIENCE

ML Researcher, Cognitive Neuroscience Mar 2024 – Jun 2025

University of Western Australia

- Built full **R pipeline** for behavioural study (n=24), from **preprocessing** → **validation**
- Modelled **2000+** reaction-time trials into cognitive parameters using **drift diffusion models**
- Separated **true impairment** from **session-order and learning confounds** using mixed-effects models; found ($p < 0.05$) elevated decision variability under impaired states

Data Analyst, Computational Psychiatry Jul 2022 – Jun 2023

IIT Kanpur

- Designed and conducted behavioural experiments with **105 human participants**; managed **recruitment, data collection, and experimental protocols**.
- Tracked learning across **10,000+ trials** using **RL models**
- Demonstrated individual learning rates explained subject variation in choice ($p < 0.05$)
- Validated findings via post-predictive checks**; presented at **national/international conferences**

Junior Research Fellow, Virology & Cell Biology Aug 2021 – May 2022

IIT Kanpur

- Ran wet-lab + imaging experiments on **SARS-CoV-2 viral entry** into mammalian cells
- Curated **MATLAB pipelines for single-particle tracking across 1000+ frames**

PROJECTS

Multi-Disease fMRI Biomarker Detection — Full-Stack Deployment Nov 2025 – Jan 2026

- Built **clinical ML system** (FastAPI + React + Docker) for schizophrenia & Parkinson’s (**n=542**)
- Improved model from **AUC 0.52** → **0.72** by identifying hidden data quality issues and applying **domain-guided feature selection** (2110 → 15 features)
- Integrated SHAP/LIME for interpretability and glass-brain visualisations.

Reverse Transcriptase Activity Prediction for Gene Editing Feb 2026 – Present

- Built **protein ML system** using **ESM-2 embeddings** + structural/motif features
- Achieved strong generalisation (**CLS=0.803, Retroviral AUC=0.833**) across **57 RT variants**
- Identified key determinants of editing efficiency (**p=0.901**)
- Ongoing **wet-lab validation to bridge prediction → biological function**

Tuberculosis Drug Resistance Prediction Aug – Dec 2020

- Modelled resistance across 3,000+ isolates, 222 SNPs using CNNs + ensemble ML
- Placed 2/10, achieving 98.4% accuracy across 10 first/second line drugs (Kaggle)

EEG Motor Imagery Classification Apr 2026

- Built full EEG processing pipeline (MNE-Python), including filtering, artifact rejection, and epoching
- Benchmarked multiple approaches (CSP+LDA, Riemannian methods), achieving up to 89% accuracy across subjects